Cancellation of n-connected maps under composition

The Univalent Foundations Program

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Editorial note. The counted statement is the author's 'lem:nconnected_postcomp' (source lines 1133-1136, Lemma 7.5.6); the proof environment starts after one blank line, so the pairing is by adjacency.

Editorial note. The retained context is the author's own corollary immediately preceding the lemma, which is the neighbouring statement in the same file and helps fix the surrounding notion.

Editorial note. The retained counter values (chapter 7, section 5, theorem offset 4) make the retained corollary print as Corollary 7.5.5 and the counted lemma as Lemma 7.5.6, matching the original book numbering.

Editorial note. The two external references are navigation pointers to the octahedron identity and to the truncation-in-a-dependent-sum theorem that the retained proof invokes by label; the counted proof is retained in full.

Editorial scope. One original result of the HoTT Book, counted once: cancellation of n-connected maps under composition (Lemma 7.5.6).

Allowed standard prerequisites (editorial).

- n-connected maps. f : A -> B is n-connected if trunc n (fiber_f b) is contractible for every b : B.. The proof compares the truncations of fibers of g and of g o f at an arbitrary c : C.
- The unstable octahedron identity. The fiber of a composite is the total space of the fibers of the second map over the fibers of the first.. It is the first equivalence in the displayed chain at source lines 1141-1143.
- Truncation inside a dependent sum. trunc n (Sigma w, trunc n P(w)) is equivalent to trunc n (Sigma w, P(w)).. It is the second equivalence at source lines 1144-1145.

Source reference. Original exercise 4.4: sources/hott-hlevels/book-labelnumbers-first-edition.tex, source lines 1414–1414. Original result cited by the retained text but located outside the retained span. It is a navigation pointer: the counted statement's own author proof is retained line for line, and no part of it is replaced by this reference.

Source reference. Original theorem 7.3.9: sources/hott-hlevels/pinned-original-source.tex, source lines 629–629. Original result cited by the retained text but

located outside the retained span. It is a navigation pointer: the counted statement's own author proof is retained line for line, and no part of it is replaced by this reference.

Corollary 7.5.5. *If g is homotopic to a n-connected function f, then g is n-connected.*

Lemma 7.5.6. *Suppose that $f : A \to B$ is n-connected. Then $g : B \to C$ is n-connected if and only if $g \circ f$ is n-connected.*

Proof. For any $c : C$, we have

$$\begin{aligned}
\|\mathsf{fib}_{g\circ f}(c)\|_n &\simeq \Big\| \sum_{w:\mathsf{fib}_g(c)} \mathsf{fib}_f(\mathsf{pr}_1 w) \Big\|_n && \text{(by original exercise 4.4)} \\
&\simeq \Big\| \sum_{w:\mathsf{fib}_g(c)} \|\mathsf{fib}_f(\mathsf{pr}_1 w)\|_n \Big\|_n && \text{(by original theorem 7.3.9)} \\
&\simeq \|\mathsf{fib}_g(c)\|_n. && \text{(since } \|\mathsf{fib}_f(\mathsf{pr}_1 w)\|_n \text{ is contractible)}
\end{aligned}$$

It follows that $\|\mathsf{fib}_g(c)\|_n$ is contractible if and only if $\|\mathsf{fib}_{g\circ f}(c)\|_n$ is contractible. □